HealthConnect Rwanda

Healthcare Operations and Performance Analytics: Business Insights Report

Data Analytics Track | Mercy Chepkemoi Koech (Miancy) | AnalystLab Africa Experience Lab | Week 6

1. Executive Summary

This report analyses 20,000 patients, 60,000 billed visits, 1,250 staff and 20 health facilities in HealthConnect Rwanda's data, using PostgreSQL (40 business queries), Power BI (a six-page dashboard) and statistical checks. **Headline numbers:** RWF 9.31bn billed, RWF 8.39bn collected (90.1%), RWF 919.9M outstanding; 74.6% of patients have a chronic condition; average satisfaction is 2.99 out of 5, with 40.4% of ratings poor (1 or 2 stars).

The three most important findings. (1) Money is being left uncollected: 19.8% of bills are short-paid, and 4,042 bills marked "Paid" still carry balances worth RWF 311.2M. (2) Half of all insurance claims (21,028, worth RWF 2.44bn) are still pending, and no claim has been rejected, which points to weak claim tracking. (3) Almost every clinical and operational measure is evenly spread across facilities, departments, diagnoses and provinces, so the data does not single out an under-performing hospital or region. The one clear pattern is the system-wide burden of chronic disease and poor patient ratings. Several results rest on a 5,000-appointment sample and a missing facility master, so they are flagged throughout (Section 5).

2. Data and Method

The supplied CSV and Excel files were loaded into PostgreSQL 18 with explicit column types and 31 tables (geography, facilities, staff, patients, clinical, billing, claims, feedback). A validation script confirmed **44 of 44 row-count and total checks** against the source files. The 40 SQL questions were each recomputed independently in Python and matched. Power BI connects to the same database through a hub-and-spoke model (Diagnoses holds one row per appointment). Comparisons were tested for statistical significance (chi-square, Kruskal-Wallis), so a difference is called a finding only when it exceeds random variation.

3. Key Business Insights

Insight 1: Revenue is not fully collected, and the Paid label cannot be trusted	
Finding	Collection is 90.1%, leaving RWF 919.9M outstanding, and the billing status field does not reflect what was paid.
Evidence	Billed RWF 9,309.4M; collected RWF 8,389.5M. 11,887 bills (19.8%) are short-paid, with an average shortfall of RWF 77,391. 4,042 of 20,145 bills marked Paid have balances worth RWF 311.2M; the outstanding total is spread evenly across Paid, Partially Paid and Pending labels (RWF 311M, 306M, 303M).
Business Meaning	Reported revenue and receivables depend on a status field that is wrong for 1 in 5 "Paid" bills. Collection effort is currently aimed at the wrong bills.
Recommended Action	Compute outstanding balances from amounts, not status. Prioritise follow-up of the 11,887 short-paid bills, starting with the largest balances, and add a rule that blocks "Paid" when payment is below the bill.

Insight 2: Half of insurance claims are pending, and none are rejected	
Finding	Claims cover most bills but are not being closed out.
Evidence	Claims exist for 70.1% of bills (42,036). 21,028 (50.0%) are Pending, worth RWF 2.44bn, and 21,008 are Approved (RWF 2.44bn). No claim is marked Rejected. Claimed value is split evenly by provider (Radiant RWF 1.24bn, Private 1.22bn, RSSB 1.21bn, MMI 1.21bn; spread under 2%). Claims total RWF 4.88bn, but only RWF 2.09bn was paid by the Insurance method.
Business Meaning	A large share of expected insurer income is unconfirmed, and with no rejection status the approval rate cannot be judged. No provider is dominant, so dependence risk is low.
Recommended Action	Set a target to resolve pending claims within a fixed number of days, record Rejected and Under Review statuses, and reconcile claims against insurer payments.

Insight 3: Chronic disease and an ageing patient group are the main clinical burden	
Finding	Three in four patients live with a chronic condition, and over a quarter are over 65.
Evidence	74.6% of patients (14,929 of 20,000) have a chronic condition: Asthma 4,996, Hypertension 4,988, Diabetes 4,945 (about 25% each). 5,487 patients (27.4%) are over 65; average age 45.0. The chronic rate differs little by province (73.5% to 76.2%; p = 0.046, a small effect). Severe cases are 33.3% of diagnoses.
Business Meaning	Long-term care, medication supply and follow-up capacity matter more than any single acute disease. The burden is national, not concentrated in one province.
Recommended Action	Create a chronic-disease management programme (scheduled follow-ups, medication adherence) and an elderly-care pathway, funded across all provinces.

Insight 4: Staffing is nearly identical everywhere, while demand is not	
Finding	Facilities and departments are staffed uniformly, regardless of workload.
Evidence	Every facility has 61 to 64 staff and every department type 125. In the 5,000-appointment sample, facilities handle 217 to 278 appointments (HF013 278, HF014 276, HF010 275 at the top; HF019 217 lowest), which is 3.5 to 4.6 appointments per staff member, a spread of about 30% (chi-square p = 0.052, suggestive rather than conclusive). Emergency visits are 33.0% of appointments. The platform has 62.5 staff per 1,000 patients and 2 nurses per doctor.
Business Meaning	Capacity is spread by headcount rather than by demand, so the busiest sites may be stretched while quieter ones have slack.
Recommended Action	Review staffing against full appointment volumes, starting with the three busiest facilities and emergency cover, before adding staff.

Insight 5: The clinical mix is even, and no disease is rising	
Finding	No diagnosis dominates and none is growing.
Evidence	Five diagnoses each account for 19.6% to 20.4% of 60,000 cases (Gastritis 20.35%, Diabetes 20.14%, Hypertension 20.12%, Malaria 19.80%, Upper Respiratory Infection 19.59%). Severity is even (33.3% severe; 32.8% to 34.1% for each diagnosis). The mix does not change by month (p = 0.32). Admission rate is 8.3%, average stay 5.49 days, referral rate 7.85%.
Business Meaning	Planning should cover all five conditions in similar proportions. The data cannot show an emerging disease trend, because only 5,000 diagnoses carry a date.
Recommended Action	Capture the appointment date on every diagnosis so trends can be tracked, and plan service lines for all five conditions in equal proportion.

Insight 6: Patient satisfaction is low across the whole system	
Finding	Satisfaction is middling and weak everywhere, and no hospital is a clear leader.
Evidence	Average rating is 2.99 out of 5: 40.4% poor (1 to 2 stars) against 19.7% excellent (5 stars). Of the 1,756 ratings linked to a facility, averages range from 2.76 (HF007, HF020) to 3.19 (HF011), but the differences are not statistically meaningful (p = 0.80, 79 to 97 ratings each). Feedback comments do not match the star ratings.
Business Meaning	Poor service experience is a system-wide issue, not one weak hospital. Facility rankings should not be used to reward or penalise sites.
Recommended Action	Launch a service-quality programme across all facilities, capture a structured reason with each low rating, and re-rank facilities once more ratings are linked.

Insight 7: Bed capacity looks ample, but occupancy is not recorded	
Finding	There is no evidence that any hospital needs more beds.
Evidence	2,000 beds (100 per facility), 400 rooms and 200 equipment items. All beds are marked Available, so utilisation is estimated from admissions: 27,438 bed-days used out of 730,000 available (3.8%).
Business Meaning	If admissions are complete, bed capacity is largely unused. If the admissions file is partial, utilisation is understated. Either way, expansion cannot be justified from this data.
Recommended Action	Record real bed occupancy and equipment status, validate the admissions volume, and decide on expansion only after that.

4. Answers to the Ten Business Questions

Question	Answer from the data	Confidence
Hospitals with most patients	HF013 (278), HF014 (276), HF010 (275) appointments in the sample; the spread is small	Low (sample)
Departments with greatest workload	Cardiology 544, Radiology 525 (sample); differences not significant (p = 0.49)	Low
Diseases becoming more common	None: five diagnoses are about 20% each with no monthly trend (p = 0.32)	Medium
Staff distributed efficiently?	Uniform (61 to 64 per facility) while demand varies about 30%; needs a demand-based review	Medium
Insurers with largest revenue	Radiant RWF 1.24bn in claims, but all four are within 2%	High
Highest patient satisfaction	HF011 (3.19), but differences are not significant (p = 0.80)	Low
Hospitals needing expansion	No evidence: estimated bed utilisation is 3.8%	Low
Reducing operating costs	No cost data. Recover RWF 919.9M outstanding and clear RWF 2.44bn pending claims	Medium
Services contributing most revenue	Visit types are near equal (Follow-up RWF 235M, Consultation 232M, Emergency 227M); Cardiology leads departments (RWF 78M). Sample covers 8.3% of collections	Low
Regions needing investment	None stands out: patients per province 3,923 to 4,091; chronic rate 73.5% to 76.2%	Medium

5. Assumptions and Data Limitations

- **Appointments are a sample.** The Appointments table has 5,000 rows (APT000001 to APT005000), but billing, diagnoses and feedback refer to appointments up to APT060000. 91.7% of bills (8.3% of billed value) cannot be linked to a facility, department, doctor or date, so facility, department and monthly analyses describe the sample only.

- **No facility master.** Facilities appear as IDs (HF001 to HF020). Names, ownership, type and location are unavailable, so province-level facility analysis is not possible.

- **Facility and department links disagree** in about 95% of staff and appointment records. The facility on the record is treated as the hospital, and the department as a department type.

- **Revenue definitions.** Revenue is payments collected (RWF 8.39bn); outstanding is billed minus paid. The Revenue Summary file totals only RWF 1.13bn and does not reconcile with billing, so it is used for the monthly trend only.

- **Proxies.** Bed utilisation is bed-days used divided by beds times 365. Revenue by service uses visit type, because no table links billing to a service. Claim approval is approved divided by all claims, because no rejected claims exist.

- **Other gaps.** Phone numbers lost digits at source (scientific notation); only 5 diagnoses exist; equipment has no availability field; patients all show Active; dates run to December 2026; there is no cost data; the data is synthetic.

Recommended data improvements: supply the full appointment tables and an official facility master, record claim rejections and bed occupancy, add failure and cost fields, and keep billing status in step with payments.